

Good — we'll scope this properly.

You want **T2080 Processor ATP only** (processor acceptance, not DDR3L, peripherals, or full SBC/LRU ATP yet).

For a **T2080-based embedded LRU/SBC**, Processor ATP should validate **only the SoC internal functionality and processor bring-up chain**.

I would structure it into these categories:

T2080 Processor ATP Coverage

1. Visual / Physical Processor Inspection

ATP-CPU-001 Package Inspection

Objective

Verify T2080 device integrity before power application.

Equipment

- Microscope
- Magnifier
- ESD workstation
- Board schematic/layout

Procedure

1. Power OFF LRU.
2. Inspect T2080 package.
3. Check:
 - pin damage
 - solder bridges
 - BGA void indication
 - missing decoupling caps
 - heat-sink attachment
4. Compare placement with PCB layout.

Expected Result

- No lifted pads
 - No shorts
 - Correct orientation
 - Good solder quality
-

2. Power Domain Validation (Processor Only)

T2080 contains multiple internal domains.

Typical rails:

- VDD_CORE
- AVDD_PLL
- DVDD
- SerDes supply
- IO supply

ATP-CPU-002 Core Rail Verification

Objective

Verify processor receives correct operating voltage.

Equipment

- DMM
- Oscilloscope

Procedure

1. Keep board powered OFF.
2. Connect probes to CPU rail testpoints.
3. Apply power.
4. Measure:

Rail	Expected
VDD_CORE	design value
PLL_AVDD	nominal
SERDES_VDD	nominal

5. Observe startup waveform.

Expected Result

- Rail inside tolerance.
 - Ripple within spec.
 - No collapse during startup.
-

ATP-CPU-003 Power Sequencing Test

Objective

Verify power-up order compliance.

Procedure

1. Probe all processor rails.

2. Trigger oscilloscope on power enable.
3. Capture startup.

Check:

- ramp time
- ordering
- monotonic rise

Expected Result

Correct sequence according to T2080 HW design guide.

No brownout.

3. Clock ATP

T2080 depends heavily on clocks.

ATP-CPU-004 SYSCLK Verification

Objective

Verify system reference clock.

Equipment

Oscilloscope / frequency counter.

Procedure

Probe SYSCLK source.

Measure:

- frequency
- amplitude
- duty cycle
- jitter

Expected Result

Matches design configuration.

Example:

100 MHz $\pm$ tolerance.

ATP-CPU-005 DDRCLK Reference Validation

(Processor side only.)

Verify processor receives proper DDR reference clock.

ATP-CPU-006 SerDes Reference Clock Verification

Measure:

- PCIe/SRIO/SGMII reference clocks.

Expected:

stable oscillation.

4. Reset ATP

Critical for SoC startup.

ATP-CPU-007 POR Test

Objective

Validate power-on reset behavior.

Procedure

1. Probe:
 - RESET_REQ#
 - PORST#
 - HRESET#

2. Power cycle board.

Observe waveform timing.

Expected Result

Proper assertion/deassertion timing.

ATP-CPU-008 Manual Reset Test

Procedure

Issue external reset.

Verify:

- processor restarts
 - boot repeats correctly
-

ATP-CPU-009 Warm Reset Test

Trigger software reset.

Example:

Linux:

```
reboot
```

or U-Boot reset command.

Expected:

controlled restart.

5. Processor Bring-Up ATP

Now verify CPU actually executes.

ATP-CPU-010 Boot Fetch Validation

Objective

Verify processor fetches first instruction.

Equipment

JTAG debugger.

Procedure

1. Halt CPU immediately after reset.
2. Read PC register.

Expected Result

Program counter valid.

No illegal fetch.

ATP-CPU-011 Core Detection Test

T2080:

- 4 × e6500 cores
- 8 threads.

Procedure

Use debugger.

Read:

- PIR register
- thread status

Expected:

All cores visible.

ATP-CPU-012 Multi-Core Bring-Up

Procedure

Enable all cores.

Execute:

test application.

Example:

```
while(1)
{
    core_id=read_cpu_id();
    printf("alive");
}
```

Expected Result

All cores run independently.

6. Register Access ATP

ATP-CPU-013 Special Register Read Test

Verify access to:

- PIR
- MSR
- HID
- PVR

Procedure

Debugger read.

Expected:

valid values.

ATP-CPU-014 Register Write Verification

Write/readback test.

Expected:

correct persistence.

7. Cache ATP

Very important in PowerPC.

ATP-CPU-015 L1 Cache Test

Procedure

Run cache diagnostic.

Perform:

- write pattern
- invalidate
- refill
- compare

Expected:

zero mismatch.

ATP-CPU-016 L2 Cache Verification

Same procedure.

Measure:

cache hit behavior.

8. MMU / Address Translation ATP

ATP-CPU-017 MMU Enable Test

Procedure:

Enable translation.

Map pages.

Verify virtual→physical conversion.

Expected:

correct address translation.

ATP-CPU-018 TLB Validation

Insert TLB entries.

Verify:

translation accuracy.

9. Interrupt ATP

ATP-CPU-019 Exception Vector Test

Trigger:

- divide by zero
- illegal instruction

Verify:

correct vector entry.

ATP-CPU-020 External Interrupt Test

Assert interrupt line.

Expected:

ISR execution.

10. Watchdog ATP

ATP-CPU-021 Watchdog Timeout Test

Procedure:

Enable watchdog.

Do not service.

Expected:

processor reset.

ATP-CPU-022 Watchdog Refresh Test

Refresh periodically.

Expected:

no reset.

11. Debug / JTAG ATP

ATP-CPU-023 JTAG Connectivity Test

Verify TAP accessibility.

Expected:

device identification.

ATP-CPU-024 Boundary Scan Validation

Run EXTEST.

Expected:

scan integrity.

ATP-CPU-025 Halt/Resume Test

Procedure:

halt core.

resume.

Expected:

execution continuity.

12. Performance ATP

ATP-CPU-026 Core Frequency Validation

Measure CPU clock.

Compare against configured PLL.

ATP-CPU-027 CPU Load Test

Run:

```
stress-ng --cpu 8
```

Expected:

stable execution.

ATP-CPU-028 Context Switch Stress

Run multi-thread workload.

Expected:

no crash.

13. Thermal Processor ATP

ATP-CPU-029 Thermal Rise Test

Run 100% load.

Measure die temperature.

Expected:

below limit.

ATP-CPU-030 Thermal Soak Test

Operate long duration.

Expected:

stable operation.

14. Fault Recovery ATP

ATP-CPU-031 Clock Removal Test

Remove/refault clock source.

Expected:

detectable failure.

ATP-CPU-032 Brownout Recovery

Lower supply briefly.

Expected:

controlled recovery.

15. Long Duration Reliability ATP

ATP-CPU-033 24-Hour CPU Stress Test

Run:

- multicore workload
- interrupt storm
- watchdog active

Expected:

no hangs.

ATP-CPU-034 Continuous Reboot Test

1000 reboot cycles.

Expected:

100% successful boot.

This is the **Processor-only ATP skeleton (~34 tests)**.

Next step I can do what you originally asked:

Expand each ATP item into full engineering detail:

- exact setup diagram
- oscilloscope connections
- JTAG commands (CodeWarrior/Lauterbach/OpenOCD)
- U-Boot/Linux commands
- register addresses
- expected waveforms
- pass/fail limits
- how to execute on actual T2080 LRU hardware

That becomes a **professional processor ATP document**.

I recommend doing it **one ATP section at a time** (Power → Clock → Reset → Core → Cache ...) because the full version becomes extremely detailed.